MODO: a Multi-GNSS and ADS-B Disciplined Oscillator with Integrity-Monitoring and Kalman Steering for Resilient Time and Frequency

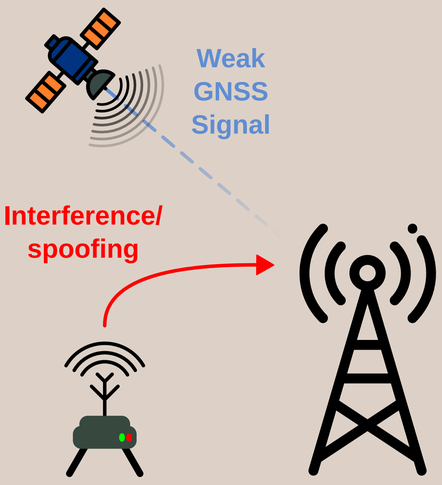
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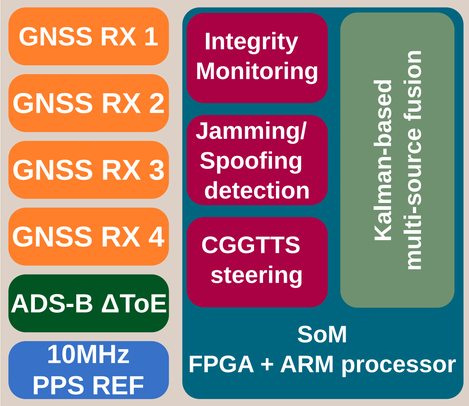
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Background - GNSS fragility



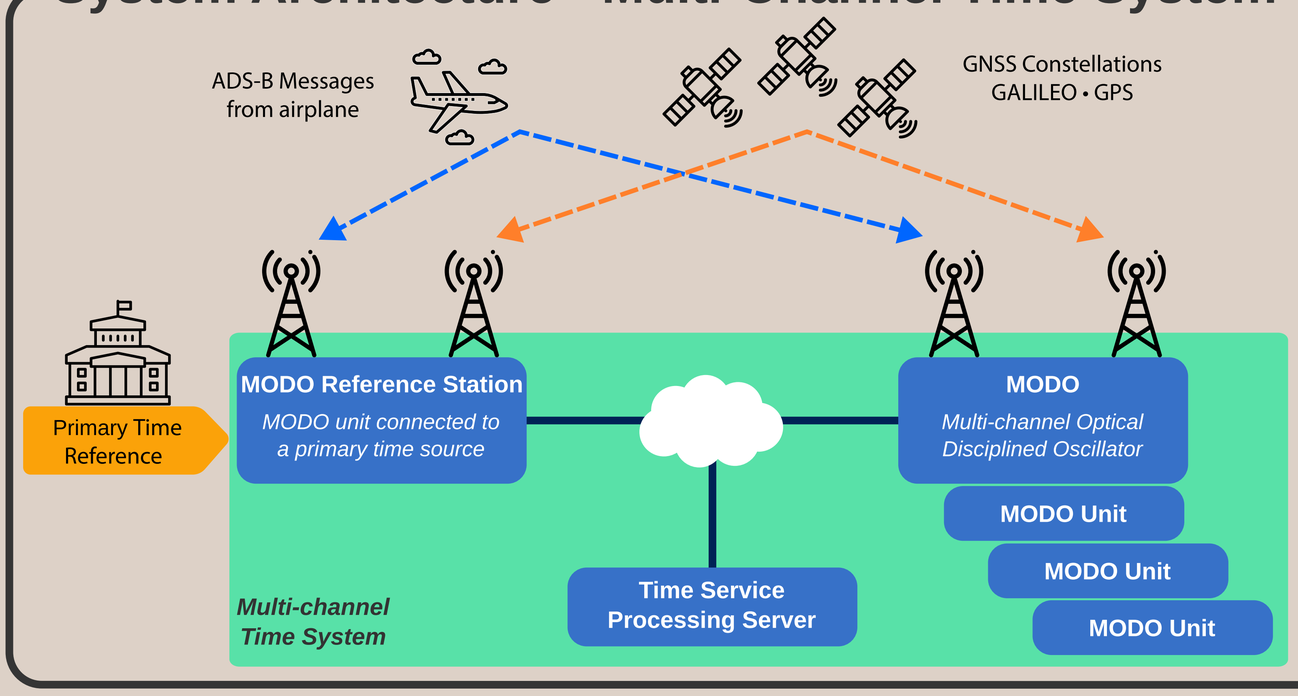
- Precise time and frequency are critical for energy, finance, telecom, and transport infrastructures.
- GNSS signals are weak (below thermal noise) and vulnerable to jamming and spoofing.
- This exposes critical infrastructures to significant resilience and security risks.

MODO Concept - multi-source time fusion



- Multi-source timing: GNSS + ADS-B + external references
- UTC-traceable reference station and distributed nodes
- Innovative TOA-based ADS-B timing observables
- CGGTTS-based steering (metrological approach)
- Network-based time dissemination (PTP/NTP)

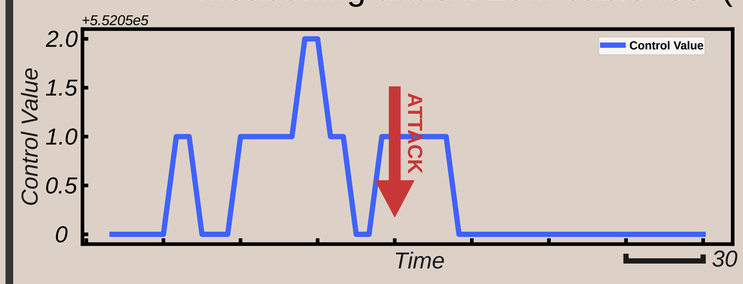
System Architecture - Multi-Channel Time System

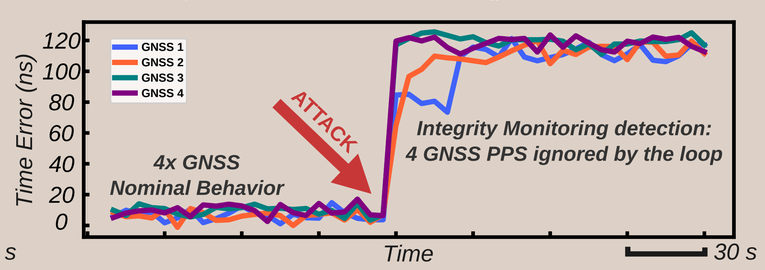


- The system integrates multi-GNSS receivers and ADS-B signals within a distributed timing architecture
- A reference station, connected to a Primary UTC source, provides traceability and calibration, including CGGTTS
- Remote MODO units operate as multi-channel disciplined oscillators, receiving and processing heterogeneous inputs
- ADS-B signals are exploited through Time of Arrival measurements, enabling independent timing information
- A central processing layer ensures data collection, synchronization, and time service distribution
- The architecture supports network-based dissemination (e.g., PTP/NTP) to multiple clients as well as local 10MHz and PPS

Results: Integrity Monitoring, Fusion and ADS-B Steering

Meaconing attack: 25m cable test (≈100ns delay added to GNSS signals)





- Robust operation under meaconing attack (~100 ns delay)
- Integrity monitor detects and excludes corrupted GNSS signals (up to 4 independent PPS)
- Stable timing maintained despite degraded inputs
- ADS-B enables independent steering for MODO devices
- Sub-ns consistency achieved on ΔTOA/ΔTOE observables
- Continuous UTC-traceable outputs preserved under GNSS threats



Meaconing attack. Top: OCXO control value and GNSS sample time series. Bottom right: Grafana screenshot showing ADS-B ΔToE values and distribution.

Key Performances

- **ADS-B TOA: σ ≈ 0.5 ns**
- **GNSS threat rejection: up to 4 channels**
- **Stability (current prototypes): ≤ 100ns**
- **Stability (KRIA-based + CGGTTS): ≤ 10 ns**

Outlook: coverage of NW Italy



A pre-operational 4-week 24/7 demo with 10 installations and ADS-B MLAT is planned in Q4 ‘26 with improved devices (AMD KRIA SoM)

Technology Roadmap

MODO Resilient Timing Unit

2025	2026	2027	2028+
Prototypes Redpitaya-based	**TRL 7** validation; **TRL 7+** AMD KRIA Platform 10-device network	**AMD KRIA-based Platform**	**Embedded** Product Development

Patent ref 102016000110784 of 03/11/2016
Method and apparatus for validating the geolocation of traceability data through open access aerospace data.

Patent ref 102019000001135 of 25/01/2019
Validation of geolocalisation and/or timing based on instants of diffusion of aircraft tracking signals.

Current Prototype

